

SWINBURNE UNIVERSITY OF TECHNOLOGY

ENVIRONMENT OPTIONS

Unit Code: Computing Technology Project A (COS40005)
Project: Internal Assistant for Enterprise Knowledge (Slack chatbot with RAG)
Client: Katalon, Inc. (replacing the original VIB client)
Team: Group 7
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A. Purpose of the Document

This document explains how Group 7 can build and test the Katalon internal-knowledge Slack assistant without access to Katalon's real Slack and Confluence. It compares **two options** for the project's test (staging) environment: **Option 1, a zero-cost environment**, and **Option 2, a low-cost environment that is closer to production**. Both options keep the architecture requested by the client: Confluence as the knowledge source, Slack as the only user interface, and Amazon Bedrock for retrieval-augmented generation.

The document is written for two readers. The **team** can use it to set up the environment and control spending. The **client mentor, Mr. Phạm Tiến Thuận Phát**, can use it to see what will be real and what will be simulated before user acceptance testing. All prices were checked on 11 November 2026 and are referenced in APA style. Prices are in US dollars and may change.

B. Production versus Test Environment

2.1 Who owns the sign-in stage

Before any employee can use Slack or Confluence, the company identity provider (IdP) decides who may sign in. In production this stage belongs to **Katalon IT**, not to the project team: HR and IT create the user in the IdP and add them to groups, SCIM provisioning creates matching accounts in Slack and Atlassian, and the employee signs in once with SAML single sign-on (Slack, n.d.-c; Okta, n.d.). The bot never decides who may enter Slack or Confluence. It only reads the email address and groups that this stage has already created.

2.2 Production components (for reference only)

The table lists what the same capabilities cost in production. These licences are owned by Katalon and are **not a project cost**. They are shown so the client can see that the test environment imitates a real, paid configuration.

Capability	Production product	List price (per user per month)	Paid by
SAML SSO and SCIM for Slack	Slack Business+	US$15, billed annually (Slack, n.d.-c)	Katalon
SAML SSO and SCIM for Confluence	Atlassian Guard Standard	US$4.20; free with Cloud Enterprise (Atlassian, n.d.-d)	Katalon
SCIM from Microsoft Entra ID (if used as IdP)	Entra ID P1	US$7, annual commitment (Redress Compliance, n.d.)	Katalon
Confluence with space and page permissions	Confluence Standard or higher	About US$5.42 for 1–10 users, annual (Featurebase, 2026)	Katalon

Because Katalon is unlikely to give an external student team access to its internal Confluence, the team needs its own test environment. Sections C and D describe the two ways to build it.

C. Option 1: Zero-Cost Environment

3.1 Components

Layer	Service	Cost	How close to production
Identity provider	Okta Integrator Free Plan	US$0	Real SAML SSO and SCIM (Okta, n.d.)
Slack	Slack Developer Sandbox	US$0 (a payment method may be required for verification only)	Real Enterprise Grid; it requires SSO, so the team must connect a real IdP (Slack, n.d.-a, n.d.-b)
Confluence	Free plan for building, then a 14-day Standard trial for permission testing and UAT	US$0	Real permissions only during the trial (Atlassian, n.d.-a, n.d.-c)
AWS	New account on the Free plan	US$0, using up to US$200 credits	Same services as production (Amazon Web Services [AWS], 2025)
Vector store	Amazon S3 Vectors	Pay per use, covered by credits	Production-grade; no minimum charge (amitrix, n.d.)
Secrets	AWS Systems Manager Parameter Store (standard)	US$0	Encrypted storage; no automatic rotation (AWS, n.d.-c)

3.2 Why the Confluence plan needs special handling

The Confluence Free plan cannot test access control. Atlassian states that the Free plan does not allow permission settings for spaces or pages (Atlassian, n.d.-c), so everyone who signs in can view and edit all content (Sharma, 2024). Because access-control correctness carries 20% to 30% of the marks, permissions must be tested on a plan that supports them. Option 1 therefore splits the work into three phases:

1. **Build phase (Free plan).** Build the ingestion pipeline, chunking, Bedrock retrieval and the Slack bot, none of which need permissions. Test the permission layer with **mock data** (for example, user A may read pages 1 and 2, user B only page 3) and unit tests, including the fail-closed rule.
2. **Integration and UAT phase (Standard trial).** Start the trial only when the bot is nearly finished. Cloud Standard can be trialled for 14 days, and a site without a card at the end of the trial is automatically moved back to the Free plan (Atlassian, n.d.-a), so no charge can occur. All access-control tests and the UAT session with the client happen in these two weeks.
3. **Fallback.** If 14 days are not enough, switch to Option 2 for one or two months.

3.3 Cost safeguards for AWS

AWS is the only service with real usage charges, so three safeguards are required.

- **Choose the Free plan at sign-up.** New customers receive US$100 in credits and can earn another US$100 by using services such as Amazon EC2 and Amazon Bedrock; the Free plan ends after six months or when credits run out, whichever comes first (AWS, 2025). No charge is made unless the account is upgraded (AWS, n.d.-a). Bedrock is not in the Always Free tier, so its usage is paid from these credits (Salinas, 2025).
- **Never use OpenSearch Serverless.** It is the quick-create default for Bedrock Knowledge Bases, but it keeps a minimum of four capacity units running and costs roughly US$700 per month even with no queries (Mallesh, n.d.), which would use all credits within about a week. S3 Vectors,

generally available since December 2025, has no minimum charge and costs nothing while idle (amitrix, n.d.).

- **Use one shared team account and do not upgrade it.** The Free plan cannot be extended beyond six months, and some account changes, such as joining AWS Organizations, upgrade it to the Paid plan immediately (AWS Builder Center, 2026). The account should be created when coding starts, not earlier.

3.4 Limitations

- Real Confluence permissions are available for only 14 days, so integration testing must be planned carefully.
- Confluence accounts are created manually with the same emails as the Slack test users, because SCIM into Atlassian requires Atlassian Guard, which is not free for students.
- The Slack sandbox allows at most eight users and two guests (Slack, n.d.-a), which is enough for the test personas but not for load testing with real users.
- Some services used by large enterprises are not available on the AWS Free plan (AWS, n.d.-a). The team must confirm on day one that Bedrock Knowledge Bases with S3 Vectors can be created.

D. Option 2: Low-Cost, Closer-to-Production Environment

4.1 What changes compared with Option 1

Option 2 keeps the same Slack sandbox, Okta IdP and AWS design. It changes only the parts where a small payment removes the main limitations of Option 1.

Layer	Option 1	Option 2	Benefit
Confluence	Free plan + 14-day trial	Standard plan for the whole project through an Atlassian Classroom licence (75% off list price)	Real space and page permissions from day one; no rush at the end
Secrets	Parameter Store (free)	AWS Secrets Manager	Automatic rotation and audit, closer to enterprise practice
AWS plan	Free plan (closes after 6 months)	Paid plan with the same credits applied first	No forced closure; full service list

Atlassian's Classroom licence is designed for collegiate and graduate students using Atlassian cloud products for learning activities and gives cloud subscriptions at 75% off list price (Atlassian, n.d.-b). On the Paid AWS plan, Free Tier credits are applied first and only usage above the credits is charged (AWS, 2025).

4.2 Estimated cost

Item	Price basis	Estimate for a 4-month project
Confluence Standard (Classroom, 75% off)	About US$680 per year for 1–10 users (Featurebase, 2026), less 75%	About US$14–17 per month, so about US$60–70
AWS Secrets Manager	US$0.40 per secret per month plus US$0.05 per 10,000 API calls (AWS, n.d.-b); one secret holds all tokens	About US$2
AWS usage (Bedrock, Lambda, DynamoDB, S3 Vectors)	Covered by up to US$200 credits if S3 Vectors and a small model are used	Expected US$0; set an AWS Budgets alert at US$10
Total		**About US$60–75, or about US$10–13 per member for six members**

4.3 Limitations

- A Classroom licence must be applied for and may be refused (Atlassian, n.d.-a); the team should apply early.
- The cost is small but not zero, and someone in the team must hold the subscription and the card.
- SCIM into Confluence still needs Atlassian Guard, so Confluence accounts are still created manually.

E. Side-by-Side Comparison

The comparison below uses the client's evaluation weights: accuracy 40%, cost and maintainability 30%, security 20% and creativity 10%.

Criterion (weight)	Option 1: Zero cost	Option 2: Low cost
Accuracy (40%)	Same bot and data; same accuracy	Same bot and data; same accuracy
Cost and maintainability (30%)	**US$0.** Strong story: S3 Vectors instead of OpenSearch Serverless saves about US$700 per month	About US$60–75 in total; still a strong story, with less time pressure
Security / access control (20%)	Real permissions tested only during a 14-day window; mock tests before that	**Real permissions tested throughout the project**; more test cycles
Creativity (10%)	Shows a disciplined, phased approach to cost	Shows production-like practice (secret rotation, longer UAT)
Main risk	The 14-day window is too short or poorly timed	Classroom licence refused or delayed
Best when	The team plans well and wants no spending at all	The team wants more time to test access control

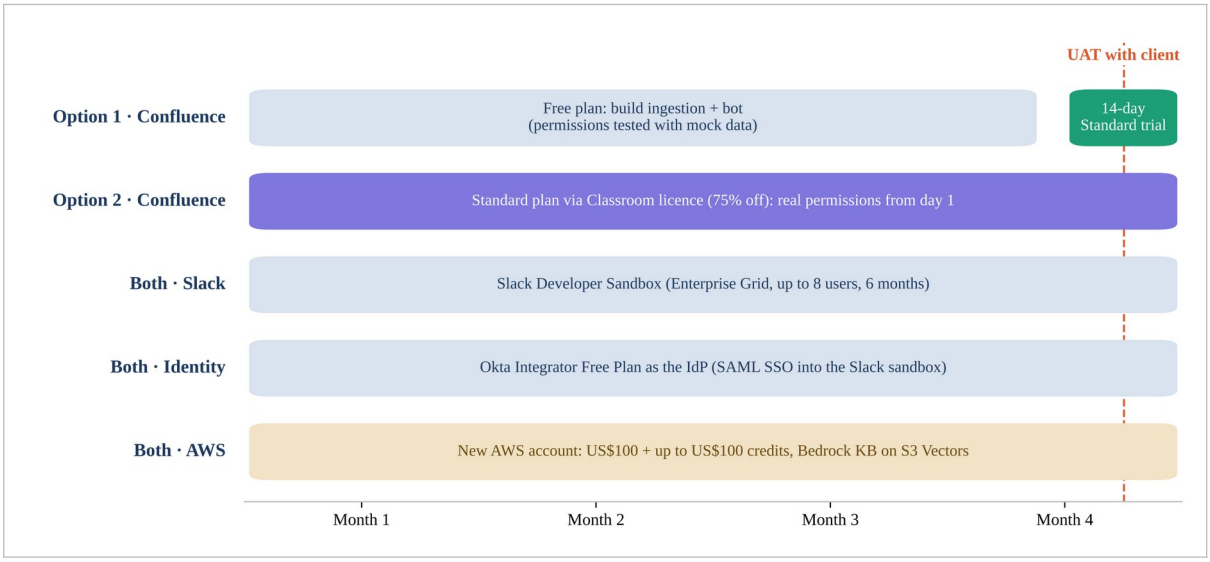


Figure 1. Timeline of both options: only the Confluence plan differs; Slack, identity and AWS are shared

F. Recommendation

Group 7 recommends **starting with Option 1** and moving to **Option 2 only if needed**. Most of the build work does not need Confluence permissions, so paying from the first day buys little. The decision points are:

1. **This week:** ask the unit convenor or Swinburne IT whether the university already holds an Atlassian academic licence. If it does, academic licence holders receive Atlassian Guard Standard and Premium at 100% off (Atlassian, n.d.-a), which would give real SSO and SCIM into Confluence at no cost.
2. **At the same time:** apply for a Classroom licence, so that Option 2 is ready if the 14-day trial proves too short.

3. **Before UAT:** start the 14-day Standard trial, or switch to the Classroom licence, at least one week before the planned UAT session with the client.

G. What Remains Different from Production

Even with the best option, a student test environment cannot match production in every way. Stating the gaps openly shows that the team understands real deployment constraints.

Aspect	Test environment	Production at Katalon	How the team compensates
Scale	Up to 8 Slack users	Hundreds of employees	Load tests with simulated requests
Data	Sample pages copying the structure of Katalon's spaces (Intranet, Deal Desk, Partner Program, True Platform)	Real internal pages	Use the page IDs and group names the client shared to build realistic samples
Provisioning into Confluence	Manual accounts with matching emails	SCIM through Atlassian Guard	The bot relies only on email and groups, so its logic is identical
Identity provider	Okta free org	Katalon's IdP (to be confirmed)	Standard SAML and SCIM, so the design transfers directly

H. Points to Confirm with the Client

1. Which identity provider does Katalon use, and is SCIM enabled for both Slack and Atlassian?
2. Is the client comfortable with UAT being run on the team's sandbox and sample pages instead of real Katalon content?
3. Does the client prefer Option 1 (zero cost) or Option 2 (about US$60–75 in total)?

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